

Multiple Choice

1. **Answer:** C

Options: A) $O(\log e^N)$; B) $O(\log N)$; C) $O(\log \log N)$; D) $O(N)$

Note: Correct option: C - $O(\log \log N)$

2. **Answer:** D

Options: A) choice(seq); B) randrange ([start,] stop [,step]); C) random(); D) seed([x])

Note: Correct option: D - seed([x])

3. **Answer:** C

Options: A) Insertion sort; B) Quicksort; C) Merge sort; D) Selection sort

Note: Correct option: C - Merge sort

4. **Answer:** C

Options: A) $O(N^{1/2})$; B) $O(N^{1/4})$; C) $O(N^{1/N})$; D) $O(N)$

Note: Correct option: C - $O(N^{1/N})$

5. **Answer:** C

Options: A) 1; B) 3; C) 8; D) 16

Note: Correct option: C - 8

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '1/K'.

Long Form Response

11. **Answer:** `def change_date_format(dt): | return re.sub(r'(\d)-(\d)-(\d)', '3-`

`2-`

`1', dt) | return change_date_format(dt)`

Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** `def count_no_of_ways(n, k): | return dp[n]`

Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key